gamma radiation Electromagnetic radiation with extremely short wavelengths (shorter than X-rays) and very high frequencies. Gamma radiation occupies the shortest-wavelength region of the spectrum. See also *electromagnetic radiation*, *electromagnetic spectrum*.

gamma-ray burst (GRB) A sudden burst of gamma radiation from a source in a distant galaxy. Gamma-ray bursts are the most powerful explosive events in the present-day universe. They may be triggered by collisions between neutron stars or black holes or by an extreme version of a supernova called a **hypernova**.

gas planet (gas giant) A large planet that, like Jupiter or Saturn, consists predominantly of hydrogen and helium. Beneath its thick gaseous atmosphere, the pressure is so great that hydrogen and helium exist in liquid form. See also *rocky planet*.

gegenschein A very faint patch of light that sometimes may be seen on a clear, moonless night in the region of sky directly opposite the position of the Sun. It is caused by sunlight that has been reflected back toward Earth by interplanetary dust particles lying beyond the orbit of Earth. See also *zodiacal light*.

general theory of relativity see *relativity*.

geocentric **(1)** Treated as being viewed from the center of Earth. **(2)** Having Earth at the center (of a system). **Geocentric coordinates** are a system of positional measurements (such as right ascension and declination) that are treated as being measured from the center of Earth. A satellite that is traveling around Earth is in a **geocentric orbit**. **Geocentric cosmology** was the ancient theory that the Sun, Moon, planets, and stars revolved around a central Earth. See also *heliocentric*.

giant star A star that is larger and much more luminous than a main-sequence star of the same surface temperature. See also *Hertzsprung–Russell diagram*, *main sequence*, *red giant*.

globular cluster A near-spherical cluster of between 10,000 and more than 1 million stars. Globular clusters, which consist of very old stars, are located predominantly in the halos of galaxies. See also *open cluster*.

gravitation see *gravity*.

gravitational lens A massive body, or a distribution of mass (such as a galaxy cluster), whose gravitational field deflects light rays from a more distant background object, thereby acting as a lens to produce a magnified or distorted image, or images, of that background object.

gravitational wave A wavelike distortion of space that propagates at the speed of light. Although waves of this kind have not yet been detected directly, there is strong indirect evidence that they exist.

gravity The attractive force that acts between material bodies, particles, and photons. According to the theory of gravity developed in the 17th century by Isaac Newton (**Newtonian gravitation**), the force of gravity acting between two bodies is proportional to the product of their masses divided by the square of the distance between their centers. For example, if the distance between the bodies is doubled, the force of attraction is reduced to one quarter of its previous value. See also *relativity*.

great circle A circle on the surface of a sphere, the plane of which passes through the center of the sphere and which exactly divides the sphere into two equal hemispheres. Its name derives from the fact that it is the largest circle that can be drawn on the surface of a sphere. See also *celestial equator*, *meridian*.

greenhouse effect The process by which atmospheric gases make the surface of a planet hotter than would be the case if the planet had no atmosphere. Incoming sunlight is absorbed at the surface of a planet and reradiated as infrared radiation, which is then absorbed by **greenhouse gases** such as carbon dioxide, water vapor, and methane. Part of this trapped radiation is reradiated back down toward the ground, so raising its temperature.

H

HII region A glowing region of ionized hydrogen surrounding one or more hot, highly luminous stars. An HII region is often just a part of a more extensive cloud of gas and dust, the remainder of which has not been ionized and is not shining. See also *ion*, *nebula*.

halo A spherical region surrounding a galaxy that contains a distribution of globular clusters, thinly scattered stars, and some gas. A **dark-matter halo** is a distribution of dark matter within which a galaxy is embedded.

heliocentric **(1)** Treated as being viewed from the center of the Sun. **(2)** Having the Sun at the center (of a system). **Heliocentric coordinates** specify the position of an object as seen from the center of the Sun. A body that is revolving round the Sun follows a **heliocentric orbit**. **Heliocentric cosmology** is a model of the universe, such as the one proposed in 1543 by Nicolaus Copernicus, in which the planets revolve around a central Sun.

heliosphere The region of space around the Sun within which the solar wind and interplanetary magnetic field are confined by the pressure of the interstellar medium. Its boundary is called the **heliopause**. See also *interstellar medium*, *solar wind*.

helium burning The generation of energy by means of fusion reactions that convert helium into carbon and oxygen. Helium burning takes place in the core of a star that has left the main sequence and become a red giant, and it may occur again, later in a star's evolution, in a shell surrounding the core. See also *fusion*, *main sequence*, *red-giant star*.

Hertzsprung–Russell (HR) diagram A diagram on which stars are plotted as points according to their luminosity and surface temperature. Luminosity (or absolute magnitude) is plotted on the vertical axis, and surface temperature (or spectral type or color) is plotted on the horizontal axis. Astrophysicists use the Hertzsprung–Russell diagram to classify stars. Depending on a star's position on the diagram, it may be classified as, for example, a main-sequence star, a giant, or a dwarf.

Hubble constant see *Hubble's law*.

Hubble's law The observed relationship between the red shifts in the spectra of remote galaxies and their distances, which implies that the speeds at which galaxies are receding are directly proportional to their distances. The **Hubble constant** (or **Hubble parameter**)—denoted by the symbol H_0—is the constant of proportionality that relates speed of recession to distance.

hydrogen burning The generation of energy by means of fusion reactions that convert hydrogen into helium. Hydrogen burning takes place in the core of a main-sequence star. When a star has consumed all the available hydrogen in its core, the core contracts and hydrogen burning then continues in a thin shell surrounding the core. See also *fusion*, *main sequence*, *proton–proton reaction*.

hypernova see *gamma-ray burst*.

I

impact crater see *crater*.

inclination The angle at which one plane is tilted relative to another. The inclination of a planetary orbit is the angle between its plane and the plane of the ecliptic (the plane of Earth's orbit). The inclination of a planet's equator is the angle between the plane of its orbit and the plane of its equator. See also *ecliptic*, *orbit*.

inferior conjunction see *conjunction*.

inferior planet A planet that travels around the Sun on an orbit that is inside the orbit of Earth. The two inferior planets are Mercury and Venus. See also *superior planet*.

inflation A sudden, short-lived episode of accelerating expansion thought to have occurred at a very early stage in the history of the universe (about 10^{-35} seconds after the beginning of time). See also *Big Bang*.

infrared radiation Electromagnetic radiation with wavelengths longer than visible light but shorter than microwaves or radio waves. Infrared is the dominant form of radiation emitted from many cool astronomical objects, such as interstellar dust clouds. See also *electromagnetic radiation*.

interstellar medium The gas and dust that permeates the space between the stars within a galaxy.

ion A particle or system of particles with a net electrical charge. Positive ions are commonly formed when an atom loses one or more of its electrons, whereas negative ions result from an excess of electrons. Ions may form from complexes of former atoms. The process by which an atom or complex gains or loses an electron to become charged is called **ionization**. See also *electron*, *photon*.

irregular cluster see *galaxy cluster*.

irregular galaxy A galaxy that has no well-defined structure or symmetry.

isotope Any one of two or more forms of a particular chemical element, the atoms of which contain the same number of protons but different numbers of neutrons. For example, helium-3 and helium-4 are isotopes of helium; a nucleus of helium-4 (the heavier, and more common, isotope) contains two protons and two neutrons, whereas a nucleus of helium-3 contains two protons and one neutron. See also *atom*, *nucleus*.

K

Kepler's laws of planetary motion Three laws, devised in the early 17th century by Johannes Kepler, that describe the orbital motion of planets around the Sun. In essence, the first law states that each planet's orbit is an ellipse, the second shows that a planet's speed varies as it travels around its orbit, and the third links its orbital period (the time taken to travel around the Sun) to its average distance from the Sun.

Kuiper Belt (Edgeworth–Kuiper Belt) A flattened distribution of icy planetesimals that orbit the Sun at distances in the region of 30–100 times Earth's distance from the Sun; it is the source of many of the shorter-period comets. See also *Oort Cloud*, *planetesimal*.

L

lenticular galaxy A galaxy that is shaped like a convex lens. It has a central bulge that merges into a disk, but no spiral arms. See also *galaxy*, *spiral galaxy*.

lepton A fundamental particle, such as an electron or a neutrino, that is not acted on by the strong nuclear force.

light-year (ly) A unit of distance equal to the distance light travels in one year—5,878 billion miles (9,460 billion km).

limb The edge of the observed disk of the Sun, the Moon, or a planet.